

CO3-AT2: LEVEL EXAM ASSESSMENT

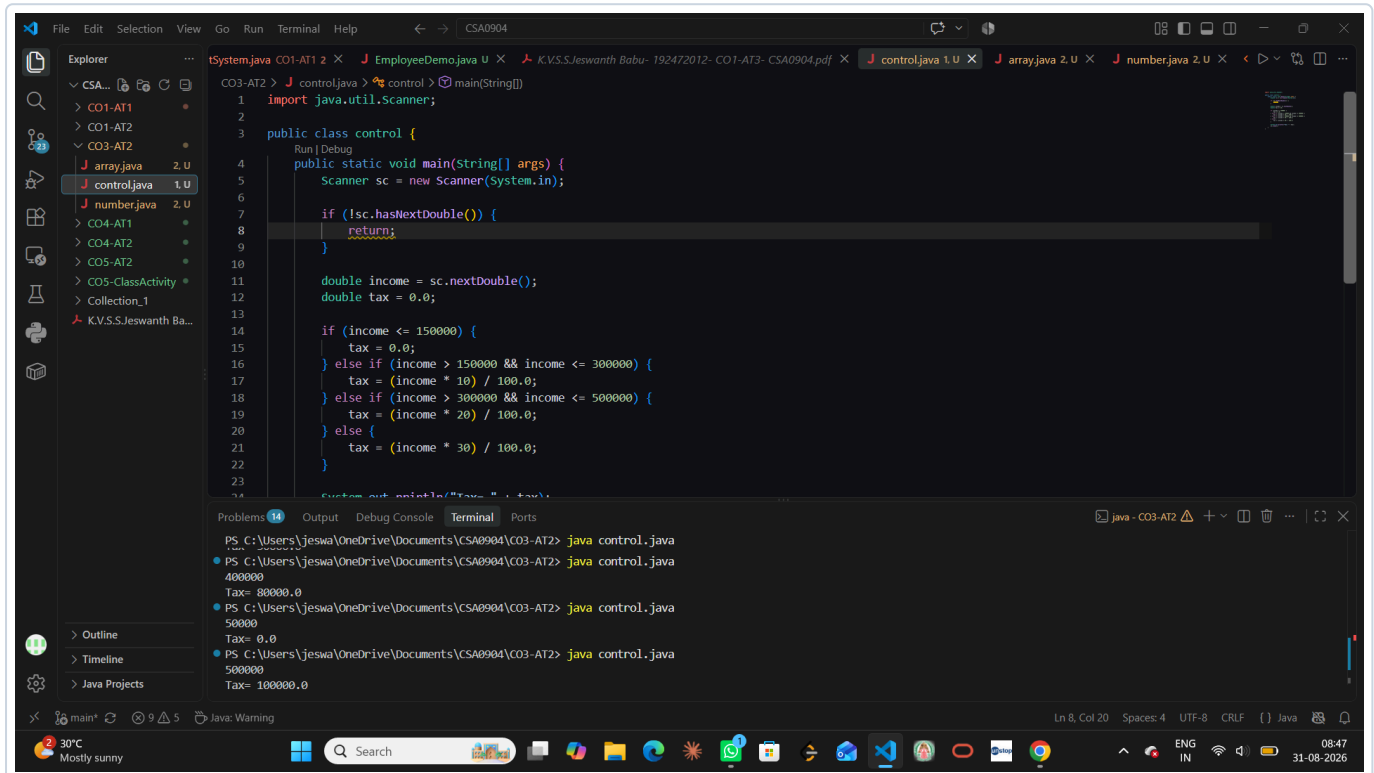
Java Programming Laboratory Documentation

Course Code	CSA0904
Course Name	Java Programming
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Department	Computer Science & Engineering (AI)
Assessment Topic	Control Structures, Array Operations & Number Theory
Date of Submission	August 31, 2026

QUESTION 1: INCOME TAX CALCULATION (CONTROL.JAVA)

Problem Statement: Write a Java program using conditional control structures (if-else if-else) to compute income tax based on the given income slabs:

- Income $\leq 1,50,000$: Tax = 0%
- $1,50,000 < \text{Income} \leq 3,00,000$: Tax = 10%
- $3,00,000 < \text{Income} \leq 5,00,000$: Tax = 20%
- Income $> 5,00,000$: Tax = 30%



The screenshot displays an IDE with a Java program named `control.java` and its execution output in the terminal.

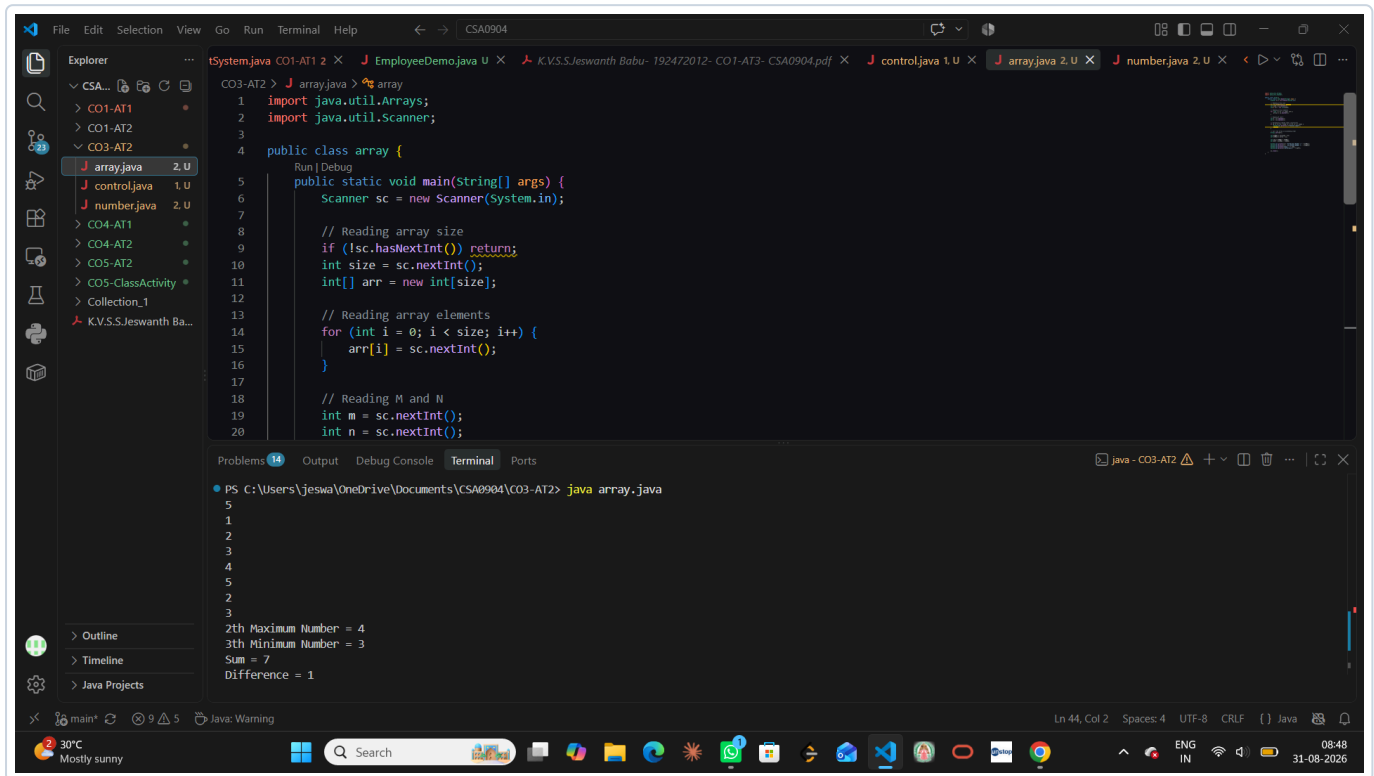
```
1 import java.util.Scanner;
2
3 public class control {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (!sc.hasNextDouble()) {
8             return;
9         }
10
11         double income = sc.nextDouble();
12         double tax = 0.0;
13
14         if (income <= 150000) {
15             tax = 0.0;
16         } else if (income > 150000 && income <= 300000) {
17             tax = (income * 10) / 100.0;
18         } else if (income > 300000 && income <= 500000) {
19             tax = (income * 20) / 100.0;
20         } else {
21             tax = (income * 30) / 100.0;
22         }
23
24         System.out.println("Tax = " + tax);
25     }
26 }
```

The terminal shows the following execution results:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO3-AT2> java control.java
400000
Tax= 80000.0
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO3-AT2> java control.java
50000
Tax= 0.0
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO3-AT2> java control.java
500000
Tax= 100000.0
```

QUESTION 2: M-TH MAXIMUM AND N-TH MINIMUM IN ARRAY (ARRAY.JAVA)

Problem Statement: Write a Java program to read an array of integers, sort the elements, and find the M^{th} maximum and N^{th} minimum numbers in the array. Compute and display their Sum and Difference.



The screenshot shows an IDE with a Java program named `array.java` and its execution output in the terminal.

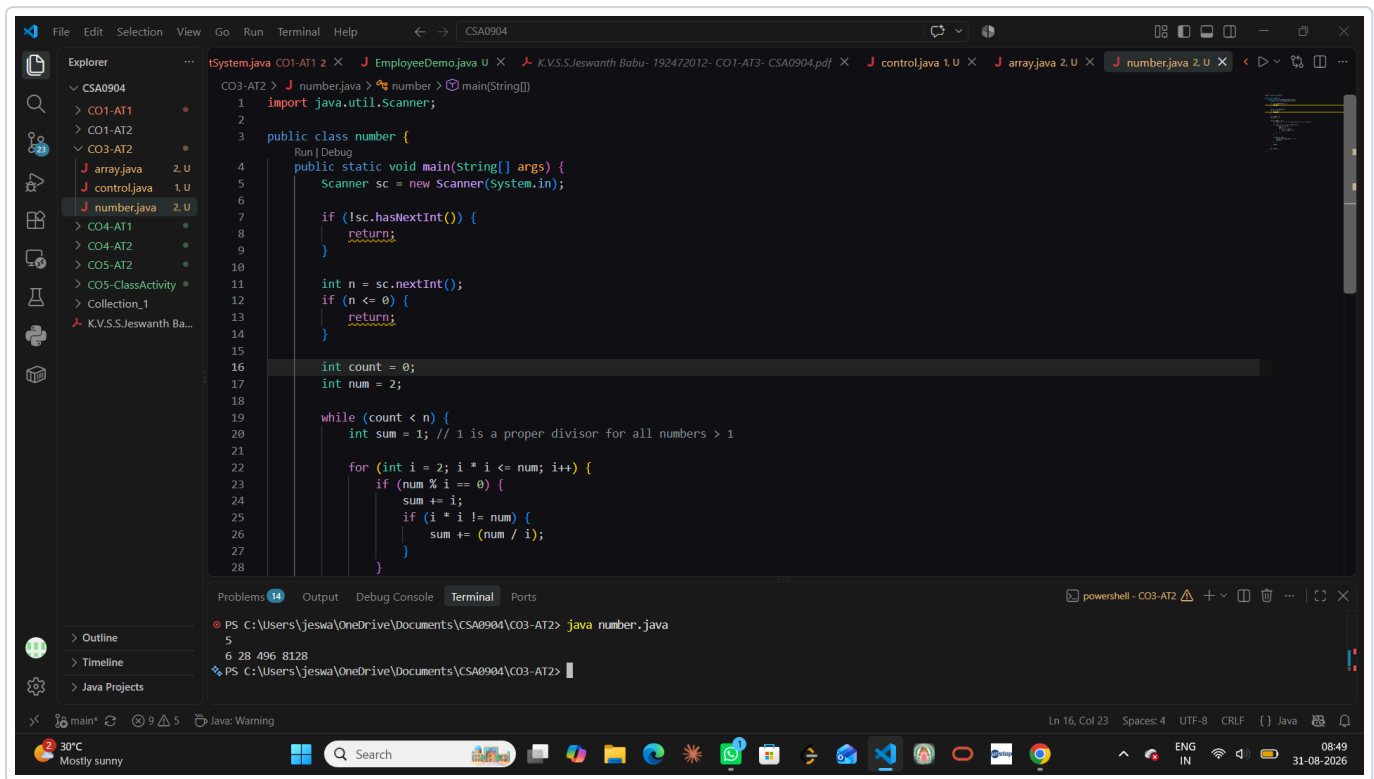
```
1 import java.util.Arrays;
2 import java.util.Scanner;
3
4 public class array {
5     Run | Debug
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8
9         // Reading array size
10        if (!sc.hasNextInt()) return;
11        int size = sc.nextInt();
12        int[] arr = new int[size];
13
14        // Reading array elements
15        for (int i = 0; i < size; i++) {
16            arr[i] = sc.nextInt();
17        }
18
19        // Reading M and N
20        int m = sc.nextInt();
21        int n = sc.nextInt();
```

The terminal output shows the program's execution results:

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO3-AT2> java array.java
5
1
2
3
4
5
2
2
3
2th Maximum Number = 4
3th Minimum Number = 3
Sum = 7
Difference = 1
```

QUESTION 3: FIRST N PERFECT NUMBERS (NUMBER.JAVA)

Problem Statement: Write a Java program to find and print the first \$N\$ perfect numbers. A perfect number is a positive integer equal to the sum of its positive proper divisors (excluding the number itself).



The screenshot shows an IDE with a Java program named `number.java` open. The program is designed to find the first `n` perfect numbers. It uses a `Scanner` to take input from the user. The logic involves a `while` loop that iterates through numbers starting from 2, and for each number, it calculates the sum of its proper divisors. If the sum equals the number, it is a perfect number, and its index and value are printed. The program then increments the count and continues the search.

```
1 import java.util.Scanner;
2
3 public class number {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         if (!sc.hasNextInt()) {
8             return;
9         }
10
11         int n = sc.nextInt();
12         if (n <= 0) {
13             return;
14         }
15
16         int count = 0;
17         int num = 2;
18
19         while (count < n) {
20             int sum = 1; // 1 is a proper divisor for all numbers > 1
21
22             for (int i = 2; i * i <= num; i++) {
23                 if (num % i == 0) {
24                     sum += i;
25                     if (i * i != num) {
26                         sum += (num / i);
27                     }
28                 }
29             }
30
31             if (sum == num) {
32                 count++;
33                 System.out.println(count + " " + num);
34                 num++;
35             } else {
36                 num++;
37             }
38         }
39     }
40 }
```

The terminal output shows the execution of the program with the input `5`, resulting in the first five perfect numbers: `6 28 496 8128`.

```
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO3-AT2> java number.java
5
6 28 496 8128
PS C:\Users\jeswa\OneDrive\Documents\CSA0904\CO3-AT2>
```